1 Mathematics for a Simple Groundwater Model

Introduction and Purpose

- Many groundwater flow models use linear algebra methods to set up and solve flow equations.
- Will use the control-volume finite-difference method as basis for forming equations describing groundwater flow.
- Purpose is to provide some insight into what is happening when using a groundwater modeling system.
- Before creating a computer model:
  - need to establish a conceptual model of system.
  - identify purpose for the computer model.

A Conceptual Model

Description
A simple 2-D model will be constructed:

- 'L' shaped aquifer
- 3000 by 4000 m
- Lake in corner, water level = 100 m
- 10 m thick fine sand over clay (elev=95 m)
- Bounded by:
  - Flow line (N)
  - Impermeable clay (E)
  - Topo. divide (S & W)

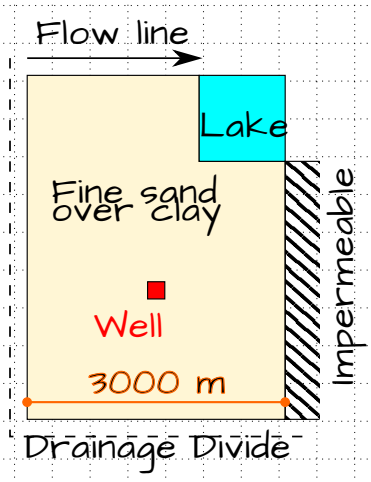

Figure 1: Conceptual model of groundwater flow

Discretizing the Model Region

To create a numerical model based on a conceptual model, the region has to be discretized or divided into discrete cells. The temporal conditions also need to be specified:

- is the model stead state or does it change with time?
- how frequently should the hydraulic head and flow rates be calculated?
- do conditions (e.g. recharge, ET) change with time?

Simple conditions will be used to construct the mathematical equations.

Choosing a Spatial Grid

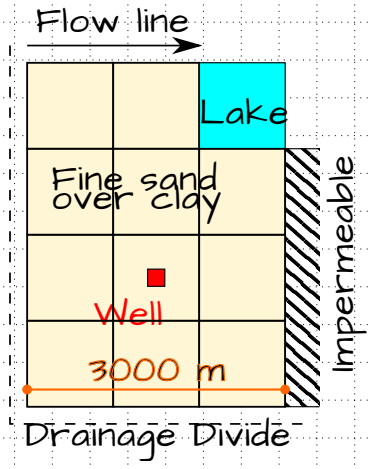

Figure 2: Simple grid for conceptual model

The grid matches the orientation of the model, minimizing the number of cells. Lake is a constant head boundary and other edges of the model are no flow boundaries.

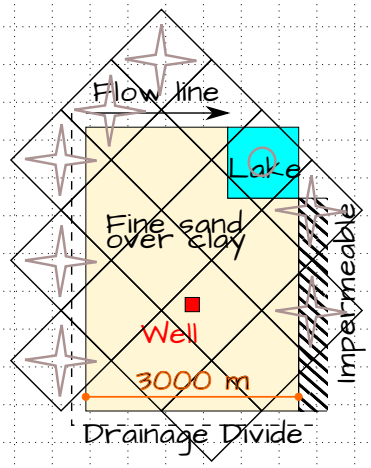

Figure 3: Alternate grid for conceptual model

Rotated grid, perhaps to align the grid axes with the hydraulic conductivity ellipse. Stars indicate no flow cells and circle indicates constant head cell.

Finite-Difference Approximations - Mass Balance

- To convert discretized conceptual model into a numerical model
  - apply numerical method (e.g. finite difference, finite elements, finite volume)
  - will use control-volume finite-difference (or finite-difference finite-volume) method
  - same method used within popular MODFLOW software

    Conservation of mass requires that the influx of mass into and out of a prescribed volume must equal the change in mass with time.

$$Q_{front} - Q_{back} = -\frac{Mass^{t+1} - Mass^{t}}{\Delta t} \tag{1}$$

Finite-Difference Approximations - Continuity Eqn

Continuity Equation

- developed using this mass balance of fluid with some density (ρ) and Divergence Theorem
- Divergence Theorem equates
  - rate that fluxes (q) diverge in region (R) of small volume (V) with a fluid content (θ).
  - flux across surface (S) of a region with

$$\int_R \nabla \cdot \rho q \, dV = \int_S \rho q \cdot n \, dS \tag{2}$$

'n' indicated normal vector

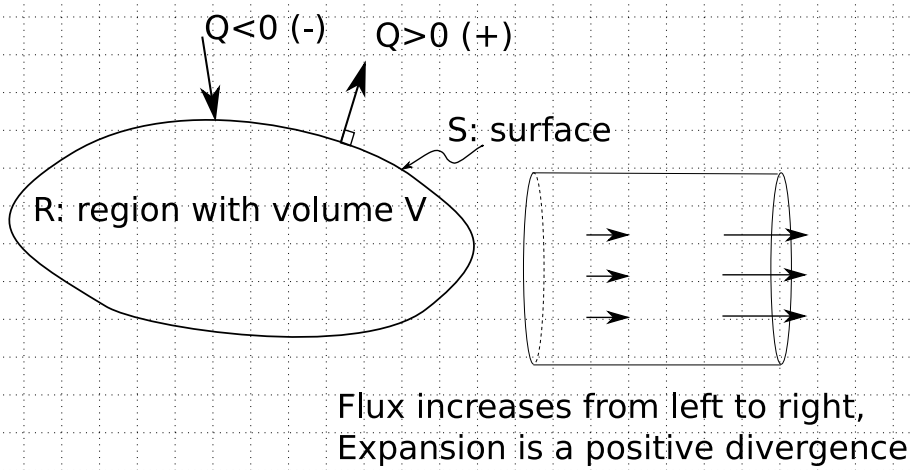

Figure 4: Divergence Theory

Finite-Difference Approximations - Continuity Eqn Cont.

Continuity Equation

- for groundwater flow, will substitute Darcy's Law for q

$$-\frac{\partial \theta}{\partial t} = \nabla \cdot (-K \nabla h) \tag{3}$$

$$\frac{\partial \theta}{\partial t} = S_s \frac{\partial h}{\partial t} \tag{4}$$

$$S_s \frac{\partial h}{\partial t} = \nabla \cdot (K \nabla h) \tag{5}$$

$$S_s \frac{\partial h}{\partial t} = \frac{\partial h}{\partial x_i} \left(K_{ij} \frac{\partial h}{\partial x_j} \right) \tag{6}$$

- x_i and x_j are x, y or z dimensions, h is hydraulic head, K is hydraulic conductivity and S_s is specific storage

To cast into equations, will use finite-difference approximations.

Finite Difference Approximations

- Based on Taylor series

$$h(x) = \sum_{n=0}^{\infty} \frac{d(h(x_0))}{dx} \frac{\Delta x^n}{n!} \tag{7}$$

$$h(x) = h(x_0) + \frac{dh(x_0)}{1!dx}(\Delta x) + \frac{d^2h(x_0)}{2!dx^2}(\Delta x)^2 \cdots \tag{8}$$

- Truncating after second term results in first order error proportional to Δx. When Δx is small, error will be small.

$$h(x + \Delta x) = h(x) + \frac{1}{1!}\left(\Delta x \frac{\partial h(x)}{\partial x}\right) + \left(\frac{\Delta x^2}{2}\frac{\partial^2 h(x)}{\partial x^2}\right) \cdots \tag{9}$$

$$\frac{\partial h}{\partial x} = \frac{h(x + \Delta x) - h(x)}{\Delta x} + \left(\frac{\Delta x}{2}\frac{\partial^2 h(x)}{\partial x^2}\right) \cdots \tag{10}$$

Finite Difference Approximations

$$h(x + \Delta x) = h(x) + \frac{\Delta x}{1}\frac{\partial h(x)}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 h(x)}{\partial x^2} + \frac{\Delta x^3}{3!}\frac{\partial^3 h(x)}{\partial x^3} \cdots \tag{11}$$

$$h(x - \Delta x) = h(x) - \frac{\Delta x}{1}\frac{\partial h(x)}{\partial x} + \frac{\Delta x^2}{2}\frac{\partial^2 h(x)}{\partial x^2} - \frac{\Delta x^3}{3!}\frac{\partial^3 h(x)}{\partial x^3} \cdots \tag{12}$$

- Adding these together yields:

$$h(x + \Delta x) + h(x - \Delta x) = 2h(x) + 2\frac{\Delta x^2}{2}\frac{\partial^2 h(x)}{\partial x^2} + 2\frac{\Delta x^4}{4!}\frac{\partial^4 h(x)}{\partial x^4} \cdots \tag{13}$$

$$\frac{h(x + \Delta x) - 2h(x) + h(x - \Delta x)}{\Delta x^2} = \frac{\partial^2 h(x)}{\partial x^2} + 2\frac{\Delta x^2}{4!}\frac{\partial^4 h(x)}{\partial x^4} \cdots \tag{14}$$

Typical Finite Difference Equations

The previously presented equations can be distilled down to:

$$\frac{\partial^2 h(x)}{\partial x^2} \approx \frac{h(x + \Delta x) - 2h(x) + h(x - \Delta x)}{\Delta x^2} \tag{15}$$

$$\frac{\partial h}{\partial x} \approx \frac{h(x + \Delta x) - h(x)}{\Delta x} \tag{16}$$

- these are approximations and errors increase with increasing value of Δx (or Δt)
- other forms of finite difference equations can be derived
- finite difference equations replace derivatives in differential equations to create discrete form

Finite-Difference Finite-Volume Method

- Based on control volumes
- Mass/volume/energy balance is performed on each cell
- Finite differences used to:
  - calculate fluxes across faces over time period
  - calculate change in storage in control volume over time period
- Flux across face between cells 1 and 2 is:

$$Q_{1,5} = -KA\frac{\partial h}{\partial x} \approx -KA\frac{h_2 - h_1}{\Delta x} \tag{17}$$

- assumes Δx is constant
- Storage changes with time (superscripts for time, subscripts for space):

$$S_s\frac{\partial h}{\partial t} = Ss_1 = \frac{h_1^1 - h_1^0}{\Delta t} \tag{18}$$

Two-Dimensional Model

- substitute FD approximations for derivatives
- assume uniform rectangular grid oriented N-S E-W, i subscripts for E-W (x) direction, j subscripts for N-S (y) direction

$$Q_E - Q_w + Q_N - Q_S = -S_s\Delta x\Delta y\Delta z\frac{\Delta h_{i,j}}{\Delta t} \tag{19}$$

$$S = S_s \cdot \Delta z \tag{20}$$

$$T_{xx} = K_{xx} \cdot \Delta z \tag{21}$$

$$\frac{T_{xx}}{\Delta x^2}(h_{i+1,j} - h_{i,j}) - \frac{T_{xx}}{\Delta x^2}(h_{i,j} - h_{i-1,j}) + \frac{T_{yy}}{\Delta y^2}(h_{i,j+1} - h_{i,j}) - \frac{T_{yy}}{\Delta y^2}(h_{i,j} - h_{i,j-1}) = S\frac{h_{i,j}^{+1} - h_{i,j}^0}{\Delta t} \tag{22}$$

- this assumes the coordinate systems of grid and hydraulic cond. are aligned

Two-Dimensional Model

$$\frac{T_{xx}}{\Delta x^2}(h_{i+1,j} - 2\cdot h_{i,j} + h_{i-1,j}) + \frac{T_{yy}}{\Delta y^2}(h_{i,j+1} - 2\cdot h_{i,j} + h_{i,j-1}) = S\frac{h_{i,j}^{+1} - h_{i,j}^0}{\Delta t} \tag{23}$$

- explicit solution: h on left side calculated at t=0 (current)
  - simpler to calculate
  - conditionally stable (need 'small' time steps)
- implicit solutions: h on left side calculated at t=1 (in future)
  - requires more linear algebra to solve

- stable

- if $\Delta x = \Delta y$ and $T_{xx} = T_{yy}$, and use implicit method

$$\frac{T}{\Delta x^2}(h^1_{i+1,j} + h^1_{i,j+1} - 4 \cdot h^1_{i,j} + h^1_{i-1,j} + h^1_{i,j-1}) = S\frac{h^{+1}_{i,j} - h^0_{i,j}}{\Delta t} \tag{24}$$

Two-Dimensional Model

Math Set-Up

- Equation 24 assigned to each cell
- Assume steady state conditions, equation's left side = 0
- Boundary conditions
  - constant head: $h_{i,j} = C$
  - fixed flux: $\frac{Q_E}{\Delta x \cdot \Delta y} = \frac{T_{xx}}{\Delta x^2}(h_{i+1,j} - h_{i,j}) = constant$
  - head-dependent: $T_{xx}\frac{h_{i,j} - h_{fixed}}{\Delta x^2}\frac{\Delta x}{\Delta x^*} = \frac{Q_E}{\Delta x \cdot \Delta y}$
    Δx^* is distance from cell center to fixed hydraulic head

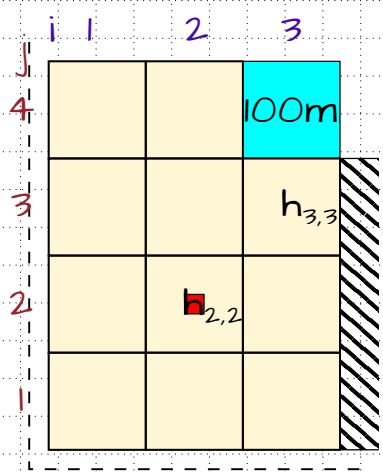

Figure 5: Grid coordinates

Setting up Equations

Equations for cells (3,1), (3,2), (2,3) with boundary conditions and source/sink terms

$$100 = h_{3,4} \tag{25}$$

$$0 = \frac{T}{\Delta x^2}[(0) - (h_{3,3} - h_{2,3}) + (h_{3,4} - h_{3,3}) - (h_{3,3} - h_{3,2})] \tag{26}$$

$$0 = \frac{T}{\Delta x^2}[(h_{3,2} - h_{2,2}) - (h_{2,2} - h_{1,2}) + (h_{2,3} - h_{2,2}) - (h_{2,2} - h_{2,1})] + \frac{Q}{\Delta x \Delta y} \tag{27}$$

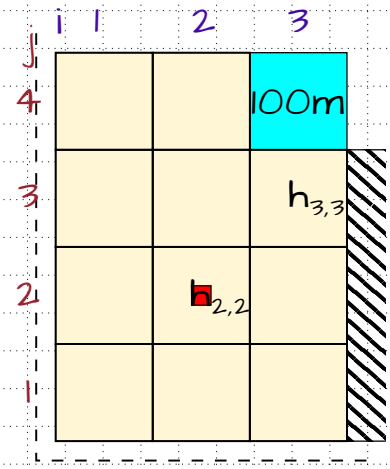

Figure 6: Grid coordinates

Equations for all cells

$$
\begin{aligned}
0 &= \frac{T}{\Delta x^2}[(h_{2,1}-h_{1,1})-(0)+(h_{1,2}-h_{1,1})-(0)] \\
0 &= \frac{T}{\Delta x^2}[(h_{2,2}-h_{1,2})-(0)+(h_{1,3}-h_{1,2})-(h_{1,2}-h_{1,1})] \\
0 &= \frac{T}{\Delta x^2}[(h_{2,3}-h_{1,3})-(0)+(h_{1,4}-h_{1,3})-(h_{1,3}-h_{1,2})] \\
0 &= \frac{T}{\Delta x^2}[(h_{2,4}-h_{1,4})-(0)+(0)-(h_{1,4}-h_{1,3})] \\
0 &= \frac{T}{\Delta x^2}[(h_{3,1}-h_{2,1})-(h_{2,1}-h_{1,1})+(h_{2,2}-h_{2,1})-(0)] \\
-\frac{Q}{\Delta x \Delta y} &= \frac{T}{\Delta x^2}[(h_{3,2}-h_{2,2})-(h_{2,2}-h_{1,2})+(h_{2,3}-h_{2,2})-(h_{2,2}-h_{2,1})] \\
0 &= \frac{T}{\Delta x^2}[(h_{3,3}-h_{2,3})-(h_{2,3}-h_{1,3})+(h_{2,4}-h_{2,3})-(h_{2,3}-h_{2,2})] \\
0 &= \frac{T}{\Delta x^2}[(h_{3,4}-h_{2,4})-(h_{2,4}-h_{1,4})+(0)-(h_{2,4}-h_{2,3})] \\
0 &= \frac{T}{\Delta x^2}[(0)-(h_{3,1}-h_{2,1})+(h_{3,2}-h_{3,1})-(0)] \\
0 &= \frac{T}{\Delta x^2}[(0)-(h_{3,2}-h_{2,2})+(h_{3,3}-h_{3,2})-(h_{3,2}-h_{3,1})] \\
0 &= \frac{T}{\Delta x^2}[(0)-(h_{3,3}-h_{2,3})+(h_{3,4}-h_{3,3})-(h_{3,3}-h_{3,2})] \\
100 &= h_{3,4}
\end{aligned}
$$

Matrix Equations

$$
\begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -\frac{Q}{\Delta x \Delta y} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 100 \end{bmatrix}
=
\begin{bmatrix}
-2 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & -3 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & -2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & -3 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & -4 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 1 & -3 & 0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 & -3 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 0 & 1 & -4 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -3 & 0 & 0 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & -2 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 & -3 & 1 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
\begin{bmatrix} h_1 \\ h_2 \\ h_3 \\ h_4 \\ h_5 \\ h_6 \\ h_7 \\ h_8 \\ h_9 \\ h_{10} \\ h_{11} \\ h_{12} \end{bmatrix}
\tag{28}
$$

Q is positive when extracting, and negative when injecting water.